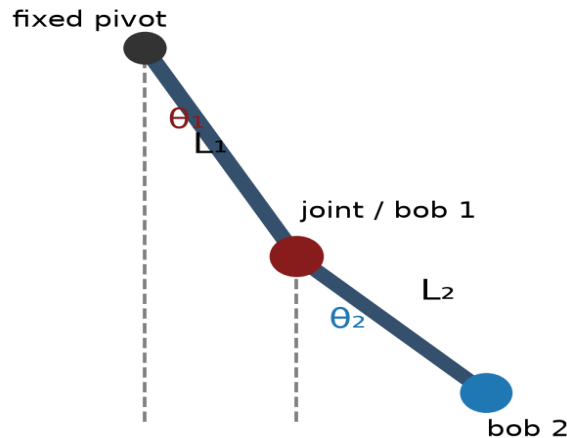


MIE 402 · Laboratory 3: Double Pendulum

Fall 2026 · Lab sections October 19–23 · Prof. Ehsan Roohi

Lab 3 and Lab 2 share a combined report, tentatively due Monday, November 2 on Canvas. Check Canvas for authoritative deadlines and section instructions. Pre-Lab 3 is due before your own Lab 3 section begins. This guide replaces Spring 2026 dates and app-version references.



Angles are absolute from downward vertical; θ_2 is not $\theta_2 - \theta_1$.

Purpose and physical picture

Measure the motion of two connected pendulum links from high-speed video. The upper link moves relative to a fixed pivot; the lower link moves relative to the upper joint. Compare measured absolute angles and their spectra with a numerical prediction based on masses and lengths you measure. Small motions can show two coupled modes; large motions can show strong sensitivity to release conditions. A few runs do not establish mathematical chaos.

Learning outcomes

- Record valid high-speed video, calibrate pixels to millimeters, and track both bob centers.
- Explain and calculate two absolute link angles and angular velocities from ExpData.
- Run a two-link model with measured bar and bob masses; compare with the experiment and explain limitations.

Equipment and required files

Double-pendulum bars, bearings, washers, nuts, bobs, balance, protractor, black background, ring light, high-speed camera with SD card, and a computer with MATLAB. Download Pendulum Motion Analyzer V4 and the Lab 3 MATLAB bundle from the course site. Preserve raw MP4 and exported MAT files. The app, not the simulation script, performs video tracking.

Quantities to measure before recording

- Upper bar mass m_1 and lower bar mass m_2 separately, in kg. Do not count joint hardware twice.

- Mass M1 attached at the upper joint and mass M2 attached to the lower bob, including assigned washers, bolts, nuts and bearings. Record what was included in each group.
- L1 from fixed pivot center to upper-joint center and L2 from joint center to lower-bob center, in meters. These are model lengths, not simply cut-bar lengths.
- Camera capture frame rate, calibration-mark distance, initial angles $\theta_1(0)$ and $\theta_2(0)$, release method and filenames.

Camera acquisition: follow this sequence

- Mount the camera securely on its tripod. Fit the ring light, use the black board, and keep cables clear. Never move the camera while recording.
- Set 1280×1024 and nominal 1000 capture frames/s, or record the actual setting used. Set trigger delay to 100% if using the course camera workflow.
- Frame the full trajectories, including large releases. Focus both bobs and the pivot; avoid bright reflections that hide the centers. Make a test swing before the actual trial.
- Start recording, release from the recorded angles without pushing, and record roughly 15 seconds or enough for the motion of interest. Stop and review.
- Mark the useful clip, save H.264 with text overlay off to SD card, wait for completion, safely remove, and copy MP4 to your storage. The playback speed shown on camera or encoded in MP4 is not the original capture rate.

Required trial design

A: a small but nonzero aligned-link release from rest. B–D: at least three larger releases with distinct initial-angle pairs; include two closely spaced starting configurations to probe sensitivity. Record actual angles instead of relying on nominal values. Keep enough frames of the entire motion for tracking. If a trial leaves the frame or becomes badly blurred, repeat it.

Analyze video in MATLAB app V4

- Install/open Pendulum Motion Analyzer V4. Select Double Pendulum, enter the original camera capture fps, and load the MP4.
- Use the known distance between the black calibration marks; the old handout uses 100 mm when the actual marks are 10 cm apart. Verify your apparatus rather than assuming.
- On the first frame, locate fixed pivot, upper bob/joint center, and lower bob center. Check the selected centers and estimated diameters.
- Run Quick Review first. Inspect both tracks. If more than 10% of points are missing, redo localization or record a better video; do not accept a broken trace.
- Analyze all frames, wait for completion, and Save Results to a MAT file. Confirm ExpData exists and has five columns; keep the raw MP4 too.



Actual Lab 3 hardware photographed in prior course material. Identify the fixed pivot, upper joint/bob and lower bob before tracking.

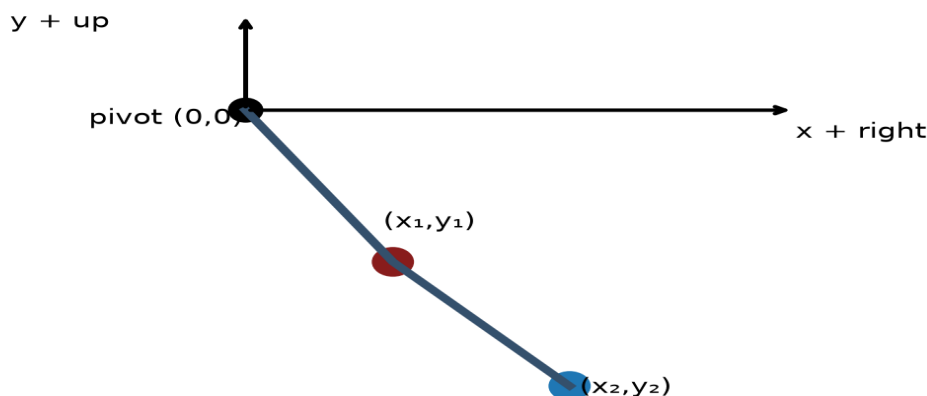
From tracked coordinates to the two angles

The V4 MAT output is $\text{ExpData} = [\text{time_s}, x1_mm, y1_mm, x2_mm, y2_mm]$. Position is relative to the fixed pivot, x positive right and y positive up. For each frame, the upper-link vector is $(x1, y1)$ and the lower-link vector is $(x2-x1, y2-y1)$. Define angles from the downward vertical:

$\theta_1 = \text{unwrap}(\text{atan2}(x1, -y1))$

$\theta_2 = \text{unwrap}(\text{atan2}(x2-x1, -(y2-y1)))$

Both should be approximately zero when both links hang downward. θ_2 is the absolute orientation of the lower link; $\theta_2 - \theta_1$ is its relative orientation. The supplied analysis script computes angular velocities by numerical differentiation and mean-removed, Hann-windowed one-sided spectra. Derivatives magnify tracking jitter; inspect spikes. The FFT requires evenly spaced samples and is not a sine fit.



Run the supplied MATLAB files

- `Run_Lab3_Simulation.m` is a runnable script. Edit measured parameters and starting conditions at its top; it calls `MIE402_Lab3_RHS.m` and `MIE402_Lab3_Energy.m`. The latter two are helper functions; do not press Run inside them.
- `Run_Lab3_ExpData_Analysis.m` is a runnable script. Select the saved MAT file; it checks columns, timing and missing coordinates, then plots the two angles, velocities and spectra. It does not call `sineFit.m`.
- `Test_Lab3_Model.m` checks a known numerical case, energy conservation of the undamped model, and the downward-vertical angle convention. Run it once before analyzing the lab files.

Model and comparison

The baseline model treats each bar as uniform and both bob/hardware groups as lumped masses. It includes gravitational torque and coupling but not friction, air resistance, flexure or out-of-plane motion. Use measured masses, lengths and initial angles. Overlay experimental and model angles for each trial, compare early timing and the main spectral content, and explain deviations honestly. For high-angle motion, long-time disagreement can arise from sensitivity to tiny initial-condition differences as well as measurement and model errors. Amplitude decay can reflect both pivot friction and air drag; do not claim one cause from a single recording.

Required evidence for the combined Labs 2–3 report

- A table listing all Lab 3 measured masses, lengths, camera capture fps, calibration distance, trial labels and actual initial angles, with units and uncertainty/measurement method.
- For every Lab 3 trial, labeled $\theta_1(t)$, $\theta_2(t)$, $\omega_1(t)$, $\omega_2(t)$, and one-sided angle spectra. A combined layout is acceptable if both links and units are legible.
- Predicted and measured angle traces compared on a common time axis; explain where they agree and diverge. Include a concise discussion of the small and large releases and the close-start pair.
- Explain how coordinates became angles, how missing tracking points were handled, the frequency resolution of the record, and why a single sine fit is not suitable for all double-pendulum traces.
- Keep raw MP4 and MAT files. Show your GTA one complete trial set (angles, velocities, spectra) before leaving; correct missing data while the apparatus is still available.

Troubleshooting and departure checklist

- App has no usable lower-bob track: verify Double Pendulum mode, center selection, focus/contrast and full-motion framing. Re-record if the bob leaves frame.
- MAT file missing ExpData: use app Save Results; do not pass the MP4 directly to the analysis script.
- “Not enough input arguments” when clicking a helper: run one of the scripts listed above, not a function file.
- Incorrect time or unexpected spectrum: verify the original camera capture fps entered in the app; the saved video playback frame rate can differ.
- Before departure: four MP4s, four MATs, trial log, measured component data, and one checked complete figure set. Do not rely on only one copy of the raw data.

Source basis: prior MIE 402 Lab 3 instructions and theory/experiment slides, updated for the Fall 2026 schedule and current V4 analyzer. The course site and Canvas govern current release and submission details.